Review class 2

Problem 1. Assume that the observation X is normal distributed with unknown mean θ and known variance 4. Assume that the prior f_θ is normal with mean 2 and variance 2. You observe $X = 2$. Calculate the posterior distribution of θ. What is the MAP estimator of θ?

Problem 2. Let x and y equal the ACT scores in social science and natural science, respectively, for a student who is applying for admission to a small liberal arts college. A sample of four students yielded the following data:

x	y
32	28
23	25
23	24
30	27

Calculate the least squares regression line for this data set. Find the ML estimator for σ^2 Find the forecast value $\hat{y}$ for $x = 22$.

Problem 3. Is my coin fair? You flip a coin 200 times. It comes up head 85 many times and tails 115 many times. Using the Chi-Square test, do you accept the hypothesis that your coin is fair at the 5% significance level? Report an interval for the p-value.

Problem 4. 6.4-8 (Hogg) Let $f(x;\theta) = (1/\theta)x^{(1-\theta)/\theta}$, $0 < x < 1$, $0 < \theta < \infty$.

(a) Find the maximum likelihood estimator of θ.

(b) Show that $\hat{\theta}$ is an unbiased estimator of θ.